

Credit Card Fraud Analytics Report

Executive Summary

- **The Business Problem:** Credit card fraud poses a significant financial risk and erodes customer trust.
- **The Approach:** I analysed a dataset of 339,607 credit card transactions over three weeks, covering exploratory data analysis, statistical hypothesis testing, and rule-based risk modelling.
- **Key Finding 1:** Transactions greater than \$500 show a fraud rate of 20.70%, far exceeding the dataset's overall average of ~0.52%.
- **Key Finding 2:** Fraud peaks during late-night hours (10 PM to 3 AM), where the fraud rate exceeds 2.6%.
- **Model Performance:** The scoring model achieved a high recall of 77.50% but low precision (2.34%), meaning it detects the vast majority of real fraud cases at the expense of a high number of false alarms.
- **Actionable Recommendation:** Implement strict verification checks for high-value purchases (> \$500) and increase monitoring on digital and late-night transactions.

Introduction

- **Background:** A new credit card company operating in the western United States aims to position itself as a secure provider.
- **The Problem:** Fraudulent transactions result in direct financial losses and poor customer experiences due to declined legitimate transactions.
- **Objective:** To identify key indicators of fraudulent transactions and evaluate a fraud detection model that prioritizes catching as many fraud cases as possible.

Data Description

- **Source & Size:** The dataset contains 339,607 credit card transactions sourced from Kaggle and adapted by DataCamp.
- **Time Period:** Covers transaction records over the analysed period.
- **Key Columns:** Transaction time, transaction amount (amt), merchant category (category), geographic location (state, latitude, longitude), and customer demographics (job, date of birth).
- **Fraud Indicator:** The target variable is is_fraud (1 = fraud, 0 = legitimate).
- **Quality Issues:** The dataset is heavily imbalanced (overall fraud rate is approximately 0.52%), which affects model performance. No major missing values were observed. Additional variables such as customer age and transaction hour were derived to aid in pattern detection.

Methodology

The analysis was conducted in three stages:

- **Week 1 (Exploration):** Data cleaning, creation of new variables (age, transaction hour), and calculation of summary statistics (mean, median, outliers).
- **Week 2 (Analysis):** Identification of fraud patterns using pivot tables, grouping (amount, category, time, state), and visualization (charts and distributions).

- **Week 3 (Modelling):** Development of a fraud scoring model, classification using thresholds, and evaluation using a confusion matrix and performance metrics (accuracy, precision, recall, and F1-score).

Key Findings

- **Finding 1: High-Value Transactions Drive Fraud**
 - **Statistic:** Transactions > \$500 → 20.70% fraud rate
 - **Chart Reference:** Amount vs. Fraud Rate
 - **Recommendation:** High-value transactions should trigger stricter verification controls.
- **Finding 2: Online Categories Have Higher Fraud Risk**
 - **Statistic:** shopping_net → 1.44%, misc_net → 1.28% fraud rates
 - **Chart Reference:** Category vs. Fraud Rate
 - **Recommendation:** Online transactions require enhanced fraud monitoring.
- **Finding 3: Fraud Peaks at Night**
 - **Statistic:** 10 PM – 3 AM → ~2.6% fraud rate
 - **Chart Reference:** Hour vs. Fraud Rate
 - **Recommendation:** Time-based monitoring rules can improve fraud detection.
- **Finding 4: Geographic Variation Exists but Is Moderate**
 - **Statistic:** Alaska → 1.69% vs. overall average 0.52%
 - **Chart Reference:** State vs. Fraud Rate
 - **Recommendation:** Location can support fraud detection but is not the primary driver.
- **Finding 5: Distance Indicates Suspicious Activity**
 - **Statistic:** Fraud transactions show higher average distance between the customer and the merchant
 - **Chart Reference:** Distance comparison
 - **Recommendation:** Geographic anomalies can signal fraud.

Model Performance

Metrics

- **Accuracy:** 82.74%
- **Precision:** 2.32%
- **Recall (Sensitivity):** 77.50%
- **F1-Score:** 4.50%

Interpretation

The model successfully identifies most fraudulent transactions, achieving a high recall rate of 82.66%. However, the extremely low precision (1.29%) indicates a high number of false alarms, meaning many legitimate transactions are incorrectly flagged as fraud.

Limitations and Assumptions

- The dataset is highly imbalanced, which affects model reliability.
- The fraud scoring model is rule-based rather than utilizing a machine learning algorithm.
- Geographic distance is approximated rather than exact.
- Results may not generalize beyond the western United States.
- Specific variables like merchant behaviour was not deeply explored.

Conclusions and Recommendations

1. **Implement Stricter Checks:** Add mandatory two-factor authentication for transactions greater than \$500.
2. **Increase Monitoring:** Strengthen rules for online transactions and late-night hours (10 PM–5 AM).
3. **Refine Thresholds:** Balance precision and recall to reduce false alarms.
4. **Distance Tracking:** Incorporate distance anomalies to catch unusual geographic activity.
5. **Consider Advanced Models:** Upgrade to machine learning classifiers in the future.

Appendix

Fraud Scoring Logic

- The fraud score is computed for each transaction using the following logic:
=IF([@amt]>500,3,0)+IF([@Amt_outliers]="Outlier",2,0) +
IF([@[Is_Night]="Night",2,0)+IF(OR([@category]="shopping_net",[@category]="misc_net"),2,0) +
IF([@[Dist_km]]>100,3,0) + IF(OR([@[Cutomer_age]]<25,[@[Cutomer_age]]>65),1,0)
- Using this scoring mechanism, predicted fraud is flagged as: =IF([@[Fraud_Score]]>=4,1,0)
- Mean Amount (Fraud): =AVERAGEIF(is_fraud, 1, amt)
- Age vs. Fraud P-value: =T.TEST(fraud_age_range, legit_age_range, 2, 3)
- Recall: =TP / (TP + FN)
- Precision: =TP / (TP + FP)

Data Dictionary

- amt: Transaction amount in USD.
- Is_Night: Categorical column indicating whether the transaction occurred at night (22:00–05:00) or during the day.
- is_fraud: Binary target variable indicating fraudulent transactions (1 = fraud, 0 = legitimate).